MCQ 1: understand and choose

Five teaching questions with a single answer. Choose one answer per question; then explain each distractor. This chapter format does not reproduce the official format of the EAM.

1. $P_A(B)$ has denominator
   - A. $P(B)$
   - B. $P(A)$
   - C. $P(A \cap B)$
   - D. $1 - P(A)$
2. $P(A) = 0{,}4,\ P_A(B) = 0{,}2$: intersection
   - A. $0{,}08$
   - B. $0{,}6$
   - C. $0{,}5$
   - D. $0{,}2$
3. A partition includes events
   - A. equal
   - B. All independent
   - C. with the same probability
   - D. disjoint and covering the sample space
4. Independence is equivalent to
   - A. $P(A) = P(B)$
   - B. $P(A \cap B) = 0$
   - C. $P(A \cap B) = P(A)P(B)$
   - D. $A = B$
5. Two disjoint events with positive probabilities are
   - A. independent
   - B. not independent
   - C. Equal
   - D. necessarily complementary

MCQ 2: understand and choose

Five teaching questions with a single answer. Choose one answer per question; then explain each distractor. This chapter format does not reproduce the official format of the EAM.

1. Three independent tosses of a fair coin, at least one tail:
   - A. $\frac{1}{8}$
   - B. $\frac{3}{8}$
   - C. $\frac{7}{8}$
   - D. $\frac{1}{2}$
2. Two independent trials with $p = \frac{1}{3}$, two successes:

A. $\frac{1}{9}$

B. $\frac{2}{3}$

C. $\frac{1}{3}$

D. $\frac{4}{9}$

3. The branches leaving the same node sum to

A. 0

B. 2

C. the probability of the path

D. 1

4. A simulation gives a frequency of 0.51. This

A. proves $p = 0{,}51$

B. is a fluctuating observation

C. proves $p = \frac{1}{2}$

D. refutes $p = \frac{1}{2}$

5. Drawing without replacement, after a red among 3 red and 2 blue, the probability of red is:

A. $\frac{3}{5}$

B. $\frac{3}{4}$

C. $\frac{1}{2}$

D. $\frac{2}{5}$

True or false? Justify.

1. $P_A(B) = P_B(A)$ for all events with positive probabilities.

2. If A and B are independent, A and $\overline{B}$ too.

3. Two successive draws are always independent.

Personal review

I find the right denominator; I check the sums at the nodes; I justify independence; I count the paths without confusing them.
For each objective, indicate: mastered without help; achieved with help; needs more work. Write an example and a mistake to avoid.